How does presentation format shape viewer response to content?

A Case Study on Educational Video Preferences



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Background

The study aimed to provide a comprehensive multimodal analysis of how the brain and body interact during exposure to naturalistic audiovisual stimuli. Reactions to three types of educational content were assessed using visual attention, neuronal activity, and cardiac patterns. These physiological measures were compared against behavioural data on liking and information retention.

26	New York, US	2026
Participants	Location	Published
69% | 31%	65% | 35%	54% | 46%
Sex	Age under 26	Student

Objectives

The analysis stands to offer several implications for the way educational content is developed, tested, and delivered.

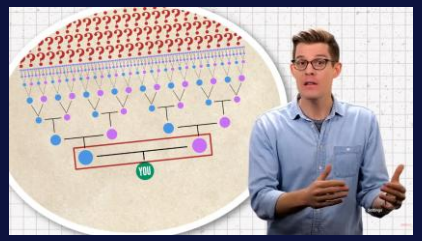
1. Determining which content formats to prioritise for maximum liking and retention.
2. Uncovering the physiological drivers behind high-performing content and paving the way for a shift from guesswork to data-backed design.
3. Revealing how audience preferences differ across segments, leading to more targeted, personalized content strategies.

Video Stimuli

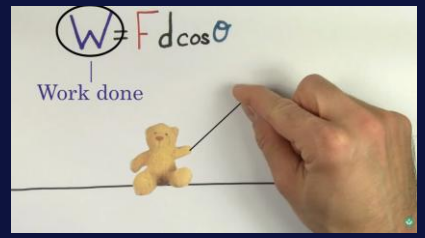
1. "What Would Happen if Mosquitoes Went Extinct?"



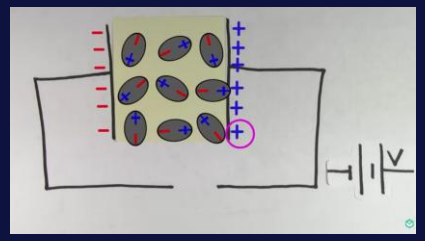
2. "Are We All Related?"



3. "Work and the Work-Energy Principle"



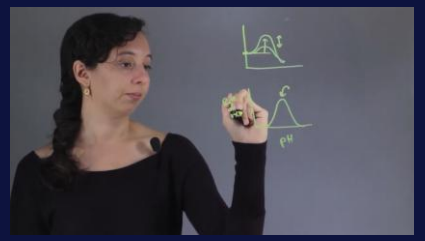
4. "Dielectrics in Capacitors; Circuits"



5. "How Do People Measure Planets and Suns?"



6. "Three Factors That May Alter the Action of an Enzyme"



Presenter and Animation **(PA)**

Writing Hand and Animation **(WA)**

Presenter and Glassboard **(PG)**

Metrics

Behavioural

Feedback
comprises ratings of each video across three attributes (Educational, Engaging, Enjoyable).

Recall
average accuracy score reflecting performance on a 10-12 question quiz assessing comprehension and retention.

Eye Tracking

Blink rate
reflects engagement and cognitive load.

Fixation rate
measures how often gaze shifts across the scene.

Average fixation duration
represents how long attention rests on points of interest.

ECG

Heart Rate
tracks changes in physiological and emotional arousal throughout the experience.

RMSSD
a measure of heart rate variability indicative of stress, autonomic regulation and cognitive effort.

EEG

Frontal Alpha Power
a measure of relaxation.

Central Theta Power
captures cognitive effort and working memory engagement.

Frontal Alpha Asymmetry
an index of left-right hemisphere balance, linked to approach versus withdrawal motivation.

Experimental Procedure



Participants were sat in front of the screen, after which EEG, ECG, and ET sensors were connected and calibrated.



One of the six videos was played, with participants instructed to pay attention to the content.



Participants shared their opinion on the video in a survey and completed a quiz assessing content retention.

Sensor Setup → Video Viewing → Feedback Survey & Recall Quiz

6x Repeated for all videos (presented in a randomised order)

Presenter and Animation videos are preferred by the viewers and support content retention most effectively

Feedback

1. **PA** content ranked first across all three scales, trailed by **PG**, with **WA** in the last place.
2. **Video 2** was the clear favourite, receiving the highest scores in every category. On the other hand, **video 3** scored lowest on Educational value, while **video 4** was perceived as the least Engaging and Enjoyable.
3. Mean-centred scores suggest that **PA** outperformed viewers' average most strongly among those under 26. Conversely, female viewers over 26 showed the smallest positive deviation from the mean – still favouring PA, but less strongly.



Recall

1. Participants scored highest on quiz questions concerning **PA** content, with moderate results for **WA**, and lowest for **PG**. There was a modest positive association between recall and liking.
2. The highest individual score was observed for **video 2**. Interestingly, despite ranking third in liking, **video 5** showed the poorest retention.
3. **PA** format boosted recall above the mean level particularly well in female non-students under 26. In males of the same group, **PA** fell below the mean. Among those viewers, **WA** format yielded the strongest recall instead.

Dynamic visuals and gesticulation win viewers, while static presentation and assembly sequences fall flat

Fixation Rate & Blink Rate

Blink Rate correlated negatively with Educational value, suggesting that videos perceived as more informative elicited greater concentration and cognitive engagement during the exposure. Fixation Rate followed similar trends but fell short of significance.

Average Fixation Duration

Here, an opposite pattern emerged, with **PG** evoking markedly longer fixations than **PA** and **WA**, which is likely linked to **PG**'s less dynamic presentation style. In fact, **video 1** – the least dynamic non-**PG** video – had the third longest fixations after **videos 5** and **6**. The signal correlated negatively with Enjoyment.



Stimuli Comparison

Video 2, which outperformed **video 1** on all scales, shows clusters of low Blink Rate and short fixation points at the start and end, coinciding with dynamic full-screen animations behind the presenter. In contrast, **video 1** used only animated text and flashing images.

Video 3, perceived as more Educational but less Enjoyable than **video 4**, gained viewers during the scene with an animated equation and a puppet, yet engagement waned when complex equations were static. For **Video 4**, attention was low during the circuit hand-building sequence but increased when the hand was writing.

Video 5 scored higher than **video 6** and won viewers over during moments when the presenter gestured extensively, contrasting **video 6**'s presenter, who moved very little.

The levels of Engagement and Enjoyment during content viewing are partly explained by autonomic relaxation

Heart Rate

The differences between both the categories and the stimuli were marginal. While no correlations with liking were observed, increased Heart Rate, reflecting elevated arousal during the viewing, was associated with lower Recall scores.

RMSSD

RMSSD showed a modest positive correlation with Engagement and Enjoyment. This points to relaxation and calm focus during exposure to favoured stimuli. Surprisingly, **video 4** had the highest RMSSD despite scoring lowest on both scales, which suggests that other factors might have driven its poor reception.



Demographic Analysis

RMSSD during **video 4** varied markedly across participant segments, indicating that the relaxation effect depended upon demographic factors. The elevated group average stemmed largely from female non-students under 26, with a smaller contribution from males over 26.

In younger female non-students, RMSSD peaked during **videos 6** and **5**, in older females – during **videos 1** and **5**. **Video 1** evoked the highest RMSSD also in young males.

Notably, in all cases, the videos with peak RMSSD were not the top behavioural performers, reinforcing that, while autonomic relaxation may contribute to content perception, it does not fully explain the differences.

Neural relaxation and mental workload measures stand out as the most reliable indicators of content success



Frontal Alpha Power

The relaxation effect was supported by neuronal data. Similarly to RMSSD, **PA** outperformed **WA** and **PG**, and showed a slight positive correlation with Engagement and Enjoyment scores. Uniquely, it was also associated with increased Recall accuracy.



Central Theta Power

The same trend held for the cognitive load marker. The metric showed the strongest ties to Feedback ratings across all signals tested, correlating positively with all three scales. Consequently, mental effort and working memory engagement emerge as the key indicators of positive content reception.



Frontal Alpha Asymmetry



The approach-avoidance index showed an unexpected pattern, with similar values across conditions and only a slight advantage for **PA**. More notably, the metric was negatively associated with Enjoyment, pointing to right hemisphere dominance and elevated withdrawal, contrary to the expected approach motivation, during exposure to liked videos.

The approach-avoidance framework may therefore not translate well to educational content assessment. Right frontal network activation may instead reflect its specialised role in controlling sustained attention and processing visual information.

Map of associations between physiological and behavioural metrics

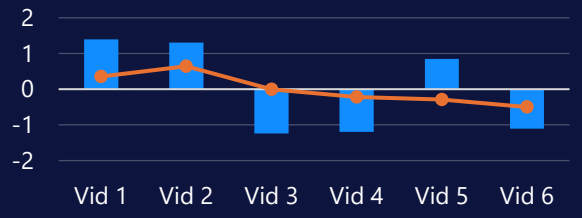
	Educational	Engaging	Enjoyable	Recall
Blink Rate	↓			
Fixation Rate				
Fixation Duration			↓	
Heart Rate				↓
RMSSD		↑	↑	
Frontal Alpha Power		↑	↑	↑
Central Theta Power	↑	↑	↑	
Frontal Alpha Asymmetry			↓	

Preliminary audience segmentation

Exploratory segmentation was conducted using hierarchical clustering in two separate analyses. The first relied on mean-centred ratings of Educational value as the segmentation variable, whereas the second was based on mean-centred ratings of Engagement and Enjoyment scores, averaged into a single variable due to high correlation between the two scales. Each analysis yielded two audience clusters. These were subsequently cross-combined into four groups (A–D), covering every pairing of cluster membership across the two analyses.
The segmentation offers a first, tentative look at demographic patterns in content preferences. A larger sample would be needed to validate these patterns and to incorporate physiological data.

Segment A

4 Participants

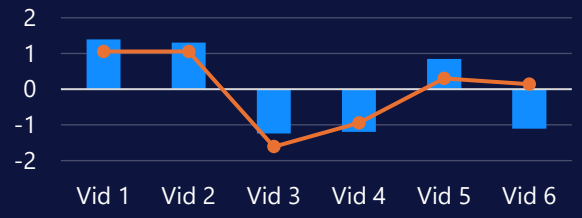


Sex	Age	Occupation
50% Male	50% Under 26	50% Student
50% Female	50% Over 26	50% Non-Student

- **PA**: Comparably high performance of both videos
- **WA**: Low Enjoyment, but moderate Educational value perceived
- **PG**: Low Education ratings, but Enjoyment of **video 5**

Segment B

7 Participants

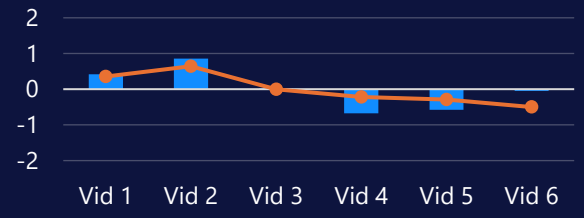


Sex	Age	Occupation
43% Male	57% Under 26	29% Student
57% Female	43% Over 26	71% Non-Student

- **PA**: High performance, especially of **video 1**
- **WA**: Most pronounced dislike of category across clusters
- **PG**: Moderate Education ratings, combined with Enjoyment of **video 5**

Segment C

10 Participants

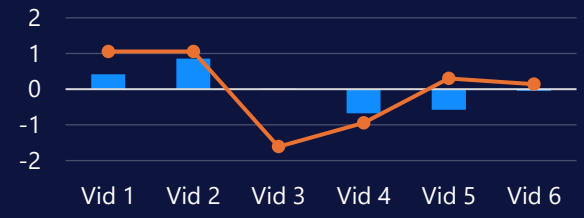


Sex	Age	Occupation
20% Male	80% Under 26	70% Student
80% Female	20% Over 26	30% Non-Student

- **PA**: Above average performance, especially of **video 2**
- **WA**: Moderate to low performance
- **PG**: Moderate to low performance

Segment D

5 Participants



Sex	Age	Occupation
20% Male	60% Under 26	60% Student
80% Female	40% Over 26	40% Non-Student

- **PA**: Above average performance, especially of **video 2**
- **WA**: Moderate to low Enjoyment and low Educational value perceived
- **PG**: Moderate to low performance

Engagement-Enjoyment Mean Education

Actionable Insights



Prioritise motion, pair presenter with animation

Among the tested content formats, Presenter and Animation videos lead as the highest-rated and most effective at conveying information. Overall, dynamic full-screen animations and an actively gesturing presenter appear to have the greatest impact on viewer appeal.



Start with the brain, confirm with heart and eyes

Successful content design evokes neural patterns of relaxation and cognitive effort, which jointly capture both subjective liking as well as knowledge transfer effectiveness. Blink Rate, Fixation Duration, and RMSSD can validate the former, while Heart Rate aids evaluation of the latter.



Be mindful of demographic factors

The study provides preliminary evidence that content preferences are shaped by sex, age, and student status. However, the small sample size and unbalanced factor distribution limit the clarity of interpretation. Further research is needed to define audience segments with greater certainty.

Next Steps

Narrowing future research down to one format, with more diverse examples, could uncover more performance drivers.

Facial Expression Analysis or talk-aloud tasks with voice analysis could reveal emotional responses to the content.

A follow-up study with a larger, balanced sample, and extended survey could strengthen audience segmentation.